

Mindsight Orchestration Pipeline Architecture

1. Architecture Blueprint

Domain ID	Domain Name	Model Architecture	Reason to Implement
M1	Personality Baseline	Grouped-Branch MLP (PyTorch)	Preserves psychometric factor boundaries via isolated parallel branches.
M2	Self-Esteem Assessment	CatBoost Regressor	Handles non-linear response reversals via symmetric tree splits.
M3	Mood & Sleep Architecture	LightGBM Classifier	Maps multi-modal feature matrix with cyclical circadian markers.
M4	Digital Dependency & Loneliness	Shared-Backbone Multi-Task MLP	Exploits clinical covariance between screen dependency and isolation.
M5	Occupational Burnout Profile	CatBoost Classifier	Matches mixed administrative features with psychological scales.
M6	Clinical Anomalies & Signs	Isolation Forest + Logistic Regression	Re-formulates clinical triage as a needle-in-a-haystack anomaly task.

2. System Constants

Global Baseline Parameter	Configured Specification Value
Orchestration Root Environment	C:\Users\Lenovo\Desktop\MINDSIGHT
Primary Demographic Pass-Through	Age (Continuous), Sex (Categorical)
Runtime Inference Format Type	Decoupled Single-Row Flat JSON
Adapter Boundary Alignment	Time-Cyclical Coordinate Mapping
Missing Evaluation Strategy	Zero-Contamination Pure-Vector Extraction
Downstream Assembly Protocol	Automated Profile Summary Aggregator